Khaled Waleed Ba Tais

AI & Computer Vision Engineer · Edge AI

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SUMMARY

Third-year Computer Science (AI) student and edge-AI builder who ships real, on-device computer-vision products — not just notebooks. Experienced across the full pipeline: dataset preparation, model training and evaluation with PyTorch and YOLO, and real-time on-device inference on Raspberry Pi 5, NVIDIA Jetson, and Hailo-8. Award-winning project team leader (Darb 2026 Winner; Top 33 of 439 teams, Asian Hackathon for Green Future 2026) seeking an AI, Computer Vision, or Edge AI internship.

EDUCATION

Taylor's University, Malaysia — dual award with University of the West of England, Bristol — Expected Aug 2027

BSc (Hons) Computer Science — Artificial Intelligence · **CGPA 3.71 / 4.00** · Focus: deep learning, computer vision, embedded systems.

TECHNICAL SKILLS

AI / ML: Python, PyTorch, TensorFlow/Keras, scikit-learn, Random Forest, XGBoost, NumPy, Pandas

Computer Vision: YOLOv5 / YOLOv8, CNNs, OpenCV, MediaPipe, object detection, segmentation, tracking

Edge AI: Raspberry Pi 5, NVIDIA Jetson, Hailo-8 (26 TOPS), ONNX, INT8 quantization, on-device / offline inference

Embedded & IoT: ESP32, Arduino, sensors, MQTT pipelines, 3D printing / CAD enclosures

Tools: Git & GitHub, Linux, Docker, REST APIs, Hugging Face Spaces, Vercel, Google Colab

PROJECTS

AquaGuard — Edge-AI Drowning Early-Warning System

Live · Code

- Built a real-time computer-vision safety system on YOLO, Raspberry Pi 5 and Hailo-8, with all inference performed locally and no cloud dependency — preserving privacy and removing network latency from the alert path.
- Trained and evaluated the vision pipeline on **8,689 images**, reaching **90.5% mAP50** for person detection and **94.2% mAP50** for pool-zone detection and segmentation.
- Implemented SAFE / DANGER / EMERGENCY decision logic with automated alerts delivered in **under 2 seconds**; validated in a real-world pool test.

Tech: Python, PyTorch, YOLO, OpenCV, Raspberry Pi 5, Hailo-8, edge deployment

ClimaShield — Edge-AI Smart Agriculture

Live · Code

- Built an autonomous rooftop-garden guardian that detects crop and leaf stress from a live camera and makes on-device decisions with a Random Forest model, auto-deploying shade when heat/moisture thresholds are crossed.
- Runs fully offline on Raspberry Pi with real-time inference; integrated sensors and ESP32/MQTT with a companion monitoring dashboard, and built a full hardware "grow-rig" prototype.
- Selected **Top 33 of 439 teams (22 countries)** for the Asian Hackathon for Green Future 2026 (For Green Future Foundation, Vingroup).

Tech: Python, PyTorch, OpenCV, scikit-learn, Raspberry Pi, ESP32, MQTT

SignBridge — Two-Way Sign-Language Translator

Demo · Live · Code

- Built a real-time translator that converts Arabic sign language into speech and speech back into text, enabling two-way communication between Deaf and hearing users.
- Shipped a public interactive demo on Hugging Face Spaces and a web app on Vercel; **winner of the Darb 2026 hackathon**.

Tech: Python, PyTorch, OpenCV, MediaPipe, Hugging Face

AWARDS & RECOGNITION

- **Darb 2026 — Winner (MYR 2,000)**; honored by the Ambassador of Yemen and the Yemeni Students Union in Malaysia.
- **Asian Hackathon for Green Future 2026 — Top 33 of 439 teams** from 22 countries; selected for the Final Round in Hanoi (For Green Future Foundation, Vingroup).
- **Registered Copyright — MyIPO (Intellectual Property Corporation of Malaysia)**, CRLY2026W05575: "ClimaShield Smart Rooftop Garden System Architecture and Operating Flow," July 2026.

LEADERSHIP & ACTIVITIES

Team Leader — led multidisciplinary teams to ship products under hackathon deadlines. Active at **AI Expo Korea (Seoul)**, ITEX 2026, and ENERtec Asia 2026 (TV interview on AI for real-world decision-making).